Rough or Just Noisy?

Source Information, Realized Volatility
and Exposure Caps

Martin Valachy
Undergraduate research working paper

22 October 2026 | Version 1.2

Abstract

This paper studies how historical trade-source information propagates into realized-volatility diagnostics and constrained portfolio inputs. A controlled comparison covers 398 historical sessions for SPY, with a fixed follow-up extending four ETFs to the same calendar: 1,990 distinct source-paired fund-days in total. For SPY, the same strict literal trade-condition filter produces materially different one-minute realized variance on 41 of 396 non-discovery dates. Finite-band slope differences do not identify latent market roughness. A fixed EWMA rule produces a persistent one-minute risk-input discrepancy, while the five-minute exposure cap suppresses 96.95% of the uncapped mean gap. Broader asset coverage qualifies that reassuring five-minute result: QQQ has 50 of 272 dependent origins with a capped gap of at least five percentage points, against 0 for SPY under the same original policy. Conversely, EFA's one-minute mean gap falls from 3.2885pp before the cap to 0.0109pp after it. Missing TLT grids withhold an incomplete-history comparison. The practical contribution is a reproducible audit separating source information, measurements, state estimates and constraints, with independent numerical checks and explicit historical/exploratory limits. It is not a new estimator, untouched-regime validation or a trading-performance claim.

Keywords: realized variance; measurement error; rough volatility; trade conditions; source compatibility; volatility targeting.
Status: complete computational working-paper draft for review; AI assistance and data availability are disclosed in Section 8. Repository distribution excludes licensed observations.

1 Introduction

Realized variance is often treated as an input to subsequent statistical and financial analysis. Before it becomes such an input, however, it is constructed from a sequence of choices: the source representation, the retained trades, the trading session, the treatment of simultaneous observations and the sampling grid. Applying the same code to two sources does not necessarily preserve the same information. A further complication arises when the resulting estimates enter a constrained decision rule. Different estimates can produce nearly identical portfolio weights if both weights reach the same cap.

This paper asks a deliberately narrow question: *how do differences in source time and condition information change measured volatility, a finite-band roughness diagnostic and a fixed risk input, when the rest of the construction is held constant?* The question is investigated using two historical WRDS TAQ representations and controlled combinations of their fields. A small known-truth simulation provides context for interpreting the roughness diagnostic; it is not presented as a new discovery.

Rough-volatility research motivates the distinction between a latent process and its observed proxy. Gatheral, Jaisson and Rosenbaum [1] provided influential evidence and modeling arguments for rough volatility. Cont and Das [2] demonstrate how realized-volatility roughness can arise under nonrough latent dynamics. Fukasawa, Takabatake and Westphal [3] show that observation error can create misleading scaling while developing a different, observation-aware inference procedure whose empirical conclusions still support roughness. These results do not license a conclusion that every low empirical slope is an artifact. They motivate checking what the statistic measures.

Source sensitivity is also established prior art. Holden and Jacobsen [4] document substantial liquidity-measure differences between Monthly and Daily TAQ. VOLARE [5] discusses source-format, timestamp and odd-lot differences in realized-measure construction. The present paper therefore does not claim to discover that timestamps, filtering or measurement error matter. Its narrower contribution is to document a controlled source-information mechanism for realized variance and quarticity, then trace that mechanism through a simple state estimate and an exposure constraint.

For SPY, the initial mechanism finding is that the frozen strict filter selects different information across the source formats. At one minute, changing flags while retaining precise time preserves much of the observed discrepancy, whereas using modern flags with legacy time produces much smaller incidence. Restoring modern category-B information largely removes the one-minute downstream weight gap under this particular pipeline. This is a conditional mechanism diagnostic, not a recommendation to remove every B-coded trade.

The practical finding is equally important: small constrained weight gaps need not imply similar risk estimates. The original five-minute mean weight gap is only 0.032 percentage points (pp), but becomes 1.0555pp before the cap. A systematic post-result sensitivity reports every fixed policy setting and separates absolute decision gaps from scale-free state comparisons. This makes the study a useful undergraduate case study in reproducible empirical finance, while keeping the economic claims modest.

The fixed four-fund follow-up makes this lesson more specific. Five-minute sampling does not eliminate the proposed-weight discrepancy for QQQ under the same original policy, while EFA exhibits substantial cap masking even at one minute. This heterogeneity is reported with every specified fund and grid, including missing comparisons, rather than used to select a favorable asset. The extension was specified after the SPY result but before the additional acquisition.

2 Data and controlled design

2.1 Sample, provenance and discovery

The original A10 sample contains every fixed XNYS session from 2 January 2013 through 31 July 2014 for SPY: 398 sessions. The fixed January 2014 panel additionally contains QQQ, IWM, EFA and TLT. Together, that original source-paired export has 482 symbol-days and 121,467,460 matched ordered input records before cleaning. These are correspondences of observed rows, not a verified count of unique exchange executions; repeated identical records can be indistinguishable.

A13 subsequently extends all four ETFs over the same 398-session calendar, adding 1,508 paired fund-days and reusing the 84 previously seen January pairs. The union with SPY contains 1,990 distinct fund-days, without counting reused observations twice. Its separate primary incidence calendar excludes the previously seen January month as well as the fixed discovery dates; it is not identical to the original SPY incidence denominator. Section 6.1 and Appendix F define and report this follow-up.

The mounted historical sources are a legacy seconds representation and a millisecond representation. Their common available overlap determines this case-study window. The source extraction

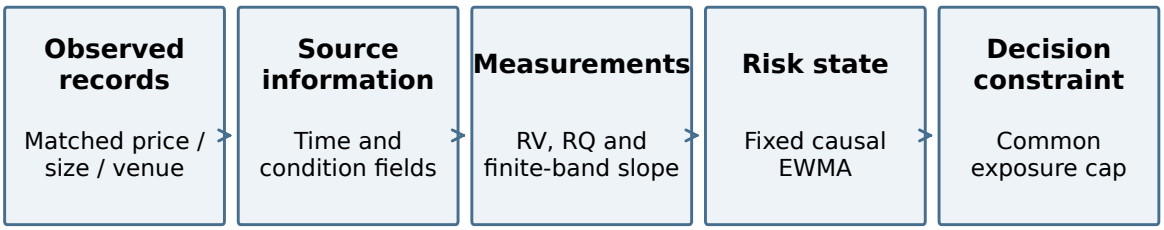


Figure 1: The observation-to-decision chain. Controlled source fields can alter measurements even when ordered observed price/size/venue records match. A common decision constraint can then conceal differences in the state estimates. The diagram describes an audit chain, not an economic causal model.

is performed on WRDS; raw ticks are not downloaded. Source subsets are checked for matching ordered price, size and venue observations before comparison, with no invented joins used to force alignment. Local exports contain daily measures, diagnostics and provenance. Matching observed records conditions the comparison on a shared observable input sequence; it does not prove that either source is a complete or error-free record of economic trades.

The mechanism was first investigated on two dates, 2 January 2014 and 3 July 2014. The expanded crossed-source design was fixed before its full run, after that discovery and earlier training exploration. Primary incidence excludes these two explicit discovery dates, leaving 396 SPY pairs at one minute. This exclusion is not advertised as an untouched chronological holdout. Full-window slope and EWMA-path calculations retain both dates. The later policy-grid exercise is explicitly post-result sensitivity. Validation observations from 2017–2019 and the final period from 2020 onward remain closed.

2.2 Five source constructions

The reference construction, denoted M, uses modern time and modern flags. Four alternatives are considered: legacy time and legacy flags (L); modern time and legacy flags; legacy time and modern flags; and legacy time/flags with modern category-B information restored (B). All retain the same remaining measurement pipeline. The restoration changes which information is available to the frozen literal filter; it is not adopted as a production cleaning rule.

The NYSE specification distinguishes condition meanings by tape: category B is average-price for CTA issues but bunched for UTP issues [6]. A single economic interpretation should therefore not be applied to all ETFs. Moreover, a source specification describes intended fields, not a proof of how a particular mounted derivative was generated. Direct floor-time and correction comparisons were additionally made on the two discovery dates; that specific stronger check is not silently generalized to every expanded-sample day.

2.3 Analytical questions and study logic

The study follows a chain of three questions. Does source representation change a measurement when the recorded price/size/venue sequence and remaining code are held fixed? Does that change alter a descriptive scaling statistic or a causal risk state? Does apparent agreement between capped inputs persist when the cap is removed? The controlled constructions address the first question; the fixed-band statistic and fixed EWMA address the second; the complete A11

matrix addresses the third. The design avoids choosing a forecasting model or policy because it happens to favor one source.

The prior expectation is a mechanism, not a desired sign: selecting different observations or changing their ordering within a grid can alter squared and fourth-power increments. Neither source must be universally larger, and smaller constrained gaps need not imply more accurate risk. Materiality thresholds define a transparent diagnostic of disagreement rather than a test of economic optimality. A10's engineering relevance criterion was fixed before its expanded analysis; the later A11 exercise explains a qualification discovered after the original results. These roles are kept distinct throughout.

3 Measurement and evaluation methods

3.1 Trading-session construction and daily moments

The fixed pipeline retains in-session positive finite price and size observations, applies its uncorrected-trade rule, and keeps only whitespace-normalized blank or @ condition strings. It selects the dominant retained venue by trade count, breaking ties by venue identifier, then takes the median price at simultaneous timestamps. Previous-tick prices are sampled on six nested grids: 15, 30, 60, 300, 900 and 1,800 seconds. The exchange calendar supplies the session length, including early closes.

There is one explicit opening convention: if no retained observation occurs exactly at the open, the opening anchor uses the first eligible trade and the approximation is flagged. Subsequent unpopulated grid points are not filled backwards from that observation. An incomplete grid retains its expected/observed counts and missing moments rather than being treated as a shorter complete session. Thus a complete recorded grid does not imply a perfectly observed economic opening price. This convention is a limitation shared across the controlled constructions.

The strict blank-or-@ rule is an intentional literal-filter diagnostic. It is narrower than the inspected `highfrequency` cleaner's condition defaults and differs in venue-selection order [7]. It is not a full replication of a general BNHLS trade-cleaning pipeline, and the results should not be interpreted as an out-of-box package comparison.

For a complete grid with M_t within-session log returns $r_{t,i}$, define

$$\mathrm{RV}_t = \sum_{i=1}^{M_t} r_{t,i}^2, \qquad \mathrm{RQ}_t = \frac{M_t}{3} \sum_{i=1}^{M_t} r_{t,i}^4. \tag{1}$$

Appendix D.1 derives both normalizations from Gaussian increment moments and explains the role of actual session counts. Appendix D.3 distinguishes price noise from daily log-proxy noise. These models give interpretive benchmarks, not assumed descriptions of the source discrepancy.

Returns are in decimal units and the moments are unannualized. The session excludes a separately constructed overnight return. A common change to percent returns multiplies RV by 100^2 and RQ by 100^4; guessed rescaling cannot resolve incompatible definitions. Numerical tolerances for independent implementation checks are separate from the materiality thresholds below.

3.2 Source discrepancy and finite-band roughness

For positive valid source pairs and a moment $Q \in \{\mathrm{RV}, \mathrm{RQ}\}$, the discrepancy is $d_{t,Q} = \log(Q_t^{\text{candidate}}/Q_t^M)$. RV is material when $|d_{t,\mathrm{RV}}| > \log(1.1)$; RQ is material when $|d_{t,\mathrm{RQ}}| > \log 2$. These are engineering thresholds, not significance tests or equivalence margins. Missing/invalid pairs do not count as agreement, and valid denominators are measure- and grid-specific.

For $x_t = \log \mathrm{RV}_t$, define the lag-k empirical increment moment and finite-band statistic

$$\widehat{V}(k) = \frac{1}{|\mathcal{I}_k|} \sum_{t \in \mathcal{I}_k} (x_t - x_{t-k})^2, \qquad \widetilde{H}_K = \tfrac{1}{2} \operatorname{slope}_{k=1,\dots,K} \{\log k, \log \widehat{V}(k)\}. \tag{2}$$

Here $\mathcal{I}_k$ retains valid endpoints on the original calendar. Missing dates never compress the lag. The primary band is $K = 10$, with $K = 5, 22$ as fixed sensitivities; paired slope differences use common endpoint support. Equation (2) is descriptive. It is not Cont and Das's normalized variation estimator or Fukasawa et al.'s quasi-likelihood estimator, and no theorem for either is transferred to it.

3.3 A fixed next-session risk input

To study propagation without selecting a forecasting model, use the initialized EWMA state and constrained weight

$$s_0 = \mathrm{RV}_0, \qquad s_t = \lambda s_{t-1} + (1-\lambda)\,\mathrm{RV}_t, \qquad u_t = \frac{v_*}{\sqrt{252 s_t}}, \qquad w_{t+1|t} = \min(c, u_t). \tag{3}$$

End-of-session t data can supply an input for $t+1$; future returns are not used. The original settings are $\lambda = 0.94$, $v_* = 0.10$ and $c = 1$. Excluding the first 126 observations leaves 272 evaluated origins. State paths require complete positive history; a gap is not imputed or removed to make the recursion run. The annualization is a scenario convention for this session-based input, not a claim to model full overnight-inclusive portfolio risk.

Appendices D.6–D.7 derive the initialized convolution, persistence of a source discrepancy, inverse-square-root sensitivity, cap threshold and scale-free source ratio. These derivations explain why one episode can affect many successive origins and why a common cap can conceal different states.

The analysis compares weight gaps, durations and input turnover. It does not evaluate returns, transaction costs, realized risk, forecast superiority or profit. A predefined practical warning criterion is met if at least 2% of valid non-discovery days are material at a grid, or at least ten post-warmup origins have an absolute weight gap of at least 5pp. The criterion classifies engineering relevance and is not a novelty test.

3.4 Uncertainty and post-result sensitivity

The incidence intervals use paired-date stationary resampling [8] with 3,999 draws. Mean block length 22 is primary; 11 and 44 are fixed sensitivities. The same resampled dates carry the numerator and valid-denominator indicators, with discovery dates ineligible for primary incidence. ETFs are not pooled as independent observations. The displayed percentile intervals describe resampling uncertainty under the specified dependence assumptions. They do not supply a uniform coverage guarantee under source-regime change, which is a material limitation of this historical window.

After seeing the original results, the A11 exercise fixes all 27 combinations of $\lambda \in \{.90, .94, .97\}$, $v_* \in \{.05, .10, .15\}$ and $c \in \{.5, 1, 2\}$. All settings and constructions are reported; no favorable policy is selected. Before capping,

$$\frac{u_t^L}{u_t^M} = \sqrt{\frac{s_t^M}{s_t^L}}, \qquad |\min(c, a) - \min(c, b)| \le |a - b|. \tag{4}$$

The uncapped ratio removes the target and common annualization convention. The second relation shows mechanically why a common cap can suppress disagreement. This sensitivity is useful but is not a new confirmatory sample or independent replication.

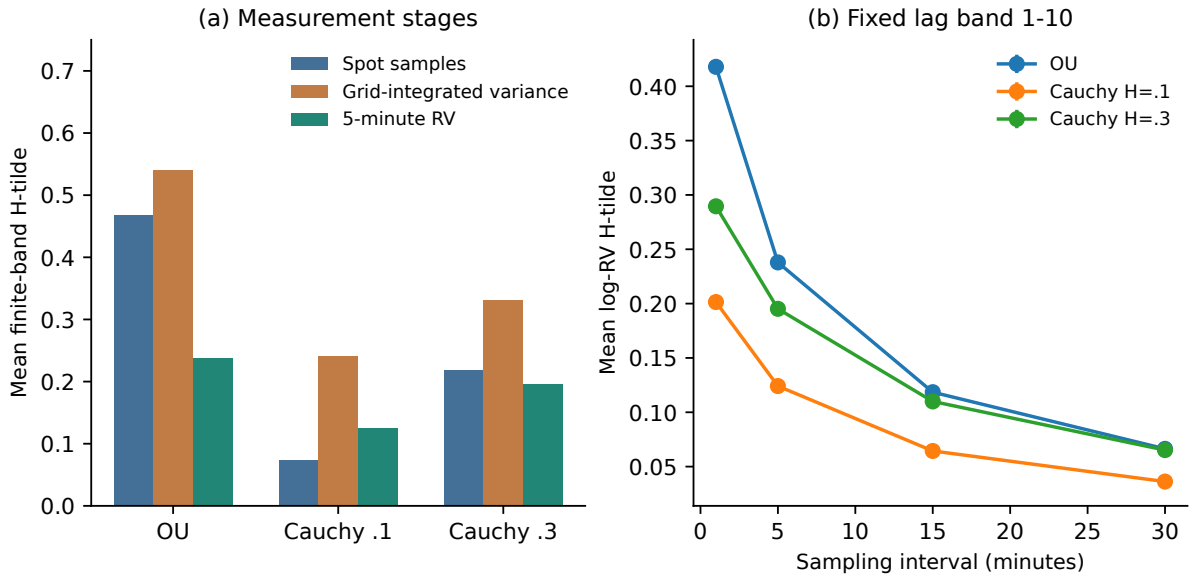


Figure 2: Known-truth simulation, no added log-price noise and calendar lags 1–10. Panel (a) compares three observation stages at the five-minute grid. Panel (b) varies the sampled-price grid; bars are ±1.96 Monte Carlo SE across independent paths, not empirical-market confidence intervals. Full noise/lag sensitivities remain in the supporting tables. Source: A08.

4 Known-truth context: a measurement diagnostic

The supporting simulation separates model regularity from the observed statistic. A stationary Gaussian log-variance process X is transformed to positive variance as $v_t = 10^{-4}\exp(X_t - 0.25/2)$. The OU case has covariance $0.25\exp(-\kappa|h|)$, with $\kappa = \log(2)/22$. Two generalized Cauchy covariance settings, following the covariance family discussed by Bennedsen, Lunde and Pakkanen [9], have local $H = .1, .3$ and the same 22-day half-correlation scale. They are stationary Gaussian Cauchy-covariance processes, not asserted to be fractional OU processes.

For each model, 128 independent paths have 504 sessions and 780 fine steps per session. Left-point variance values generate conditionally Gaussian log-price increments $\sqrt{v_j/780}\,Z_j$. Independent log-price noise of standard deviation 0, 1 or 3 basis points is added at fine-grid price endpoints; nested coarse returns sum the resulting increments. Daily grid-integrated variance is the mean of the fine variance values. Sampling intervals are 1, 5, 15 and 30 minutes, with the same fixed lag bands used for interpretation. The simulator is deliberately controlled and not calibrated to a full market microstructure process.

In the no-added-noise OU setting, local $H = .5$, while mean finite-band slopes from daily spot samples, grid-integrated variance and five-minute RV are respectively 0.467026, 0.539210 and 0.237906. The Monte Carlo SE of the last statistic is 0.002572. Figure 2 shows the observation-stage and grid dependence. This is an implementation-backed explanatory example, consistent with established concerns about proxy roughness [2, 3]; it does not establish the latent process behind SPY or refute rough-volatility models.

Table 1: Full SPY grid comparison: legacy versus modern. Incidence excludes the two discovery dates. H-tilde uses full-window calendar lags 1–10. The mean weight gap and large-gap origins use the original fixed policy and 272 evaluated origins.

Grid	RV material	RQ material	$\widetilde{H}_M$	$\widetilde{H}_L$	Mean gap (pp)	$\geq$ 5pp origins
15s	137/396	127/396	0.1703	0.0828	9.225	130
30s	75/396	73/396	0.1629	0.1228	3.261	49
1m	41/396	37/396	0.1542	0.1239	2.632	47
5m	19/396	5/396	0.1409	0.1409	0.032	0
15m	29/396	4/396	0.1046	0.1014	0.010	0
30m	51/396	5/396	0.0936	0.0912	0.014	0

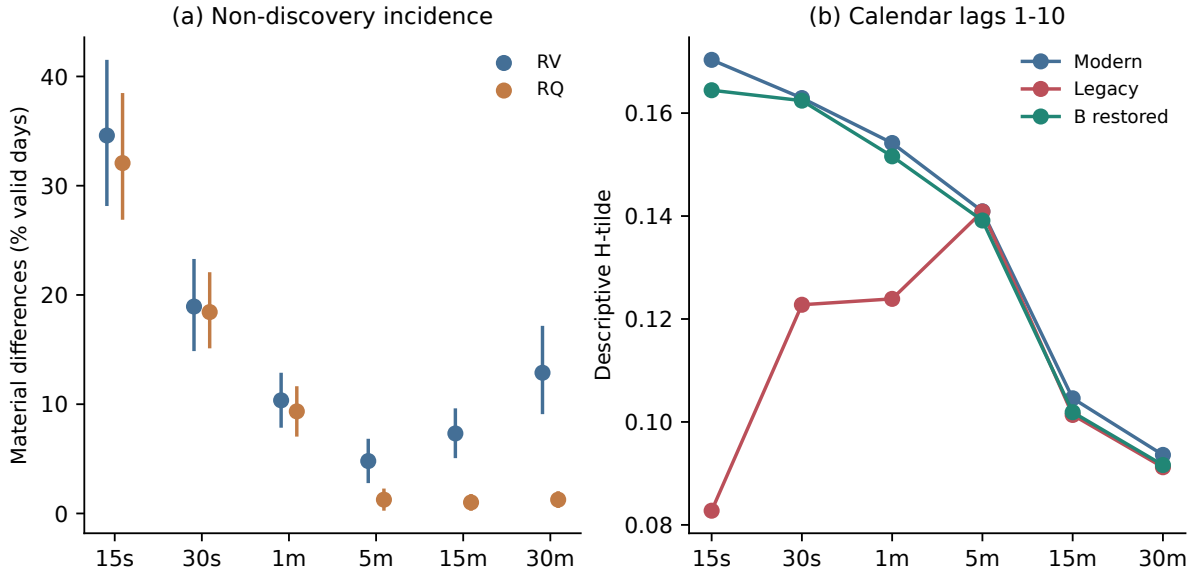


Figure 3: Source information and measured volatility. Panel (a) shows material RV/RQ incidence over valid non-discovery dates, with primary mean-block-22 percentile intervals. Panel (b) shows descriptive full-window H-tilde, including discovery dates. Different denominators/interpretations are intentional and must not be conflated. Source: A10.

5 Historical source-comparison results

5.1 Material discrepancies and descriptive slopes

Table 1 reports every fixed grid. At one minute, legacy versus modern RV is material on 41/396 non-discovery dates and RQ on 37/396. The primary block-resampling intervals for their incidence are 7.85–12.88% and 7.04–11.65%, respectively. Materiality is not monotone in grid width: coarser sampling does not eliminate every daily discrepancy. The five-minute comparison has 19 material RV dates but substantially fewer RQ discrepancies.

The one-minute finite-band slopes are 0.1542 for modern information and 0.1239 for legacy information; B restoration gives 0.1516. These are different descriptive statistics on the same underlying observed-row sequence, not estimates with reported latent-H confidence intervals. Figure 3 puts the source incidence and slope effects together. Appendix C retains the other lag bands so the interpretation does not rely on a selected slope.

Table 2: One-minute controlled constructions versus the modern reference. All incidence denominators are 396; discovery dates are excluded. The modern reference compared with itself has zero discrepancy by construction.

Construction	RV material	RQ material	Mean abs. log RV ratio
Legacy time / legacy flags	41	37	0.08524
Modern time / legacy flags	42	35	0.08500
Legacy time / modern flags	5	4	0.02562
Legacy + B information	9	5	0.02744

5.2 What the crossed constructions identify

The crossed cases in Table 2 support a limited information-mechanism interpretation. At one minute, retaining legacy flags while restoring precise time still produces 42 material RV dates. Using modern flags with legacy time reduces the count to 5; B restoration under legacy time/other flags gives 9. The comparison isolates which input information drives discrepancies under the frozen pipeline. It does not prove that flags alone have an additive causal contribution: conditioning changes can change the dominant venue and timestamp medians too.

The short fixed ETF panel provides a limited check of breadth. Excluding the January discovery date, material one-minute RV counts are 4/20 for EFA, 1/20 for IWM, 2/20 for QQQ and 0/20 for TLT; corresponding RQ counts are 3/20, 1/20, 2/20 and 1/20. These correlated single-month outcomes are not pooled as independent market-wide replications. They are reported to prevent the principal SPY result from being mistaken for a universal statement.

5.3 Propagation into a risk input

Under the original policy, the one-minute legacy/reference mean absolute weight gap is 2.632pp and the maximum is 23.9136pp. At least a 5pp difference occurs on 47 of 272 origins. Those origins form a *single consecutive episode*, from 23 January through 31 March 2014; they are not 47 independent new shocks. Figure 4 shows the path comparison. B restoration reduces the mean one-minute gap to 0.0124pp under this policy.

An existing post-result diagnostic standardizes only the two discovery-day legacy RV effects to the modern values, without deleting dates or compressing the calendar. It still leaves 46 origins with at least a 5pp gap. Omitting those dates' increment endpoints gives one-minute slopes of approximately 0.1564 modern and 0.1353 legacy. The full-window result is therefore not solely the two explicit discovery observations, but this sensitivity is not an untouched confirmation or an adopted outlier-cleaning rule.

6 Why exposure agreement can be misleading

The original five-minute constrained mean gap is 0.032pp. Before capping, it is 1.0555pp at the same target. Both sources are capped on 95.59% of evaluated origins, and the cap reduces the mean absolute discrepancy by 96.95%. Even at one minute the cap attenuates the mean gap by 70.38%. Equation (4) makes this result mechanically understandable: both inputs can be mapped to the same constrained weight despite different states.

This qualification does not erase the observed grid difference. Across the three fixed decays, one-minute uncapped weight ratios exceed a 1.1 factor on 64–77 of 272 origins. Five-minute counts are zero. The transformed p95 comparisons in Figure 5 use uncapped ratios, independent of the target, and remain smaller at five minutes. These are relative comparisons of two estimated

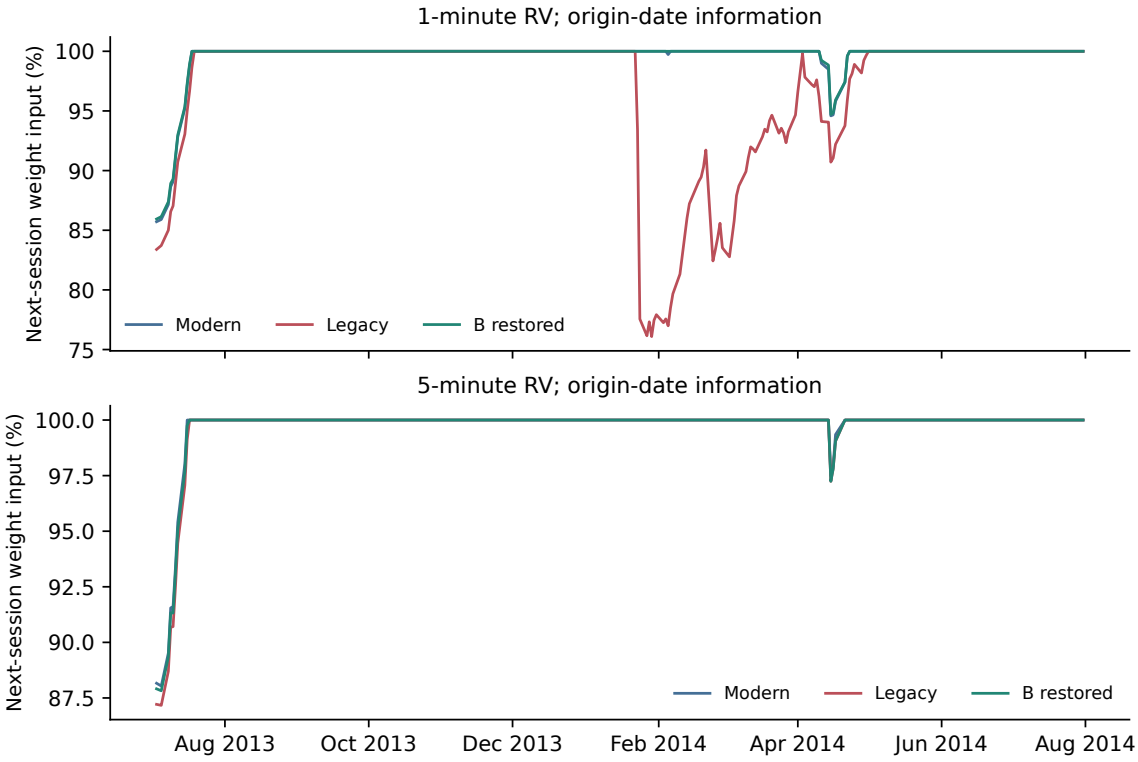


Figure 4: Fixed policy weight inputs, labeled by the end-of-session origin date that supplies information for the next session. Settings are $\lambda = .94$, target 10%, cap 100% and warmup 126 observations. The source-dependent one-minute episode is persistent. Five-minute constrained paths appear closer; Figure 5 explains why that cannot alone certify similar states. No portfolio returns are evaluated. Source: A10; displayed paths are checked against its summaries.

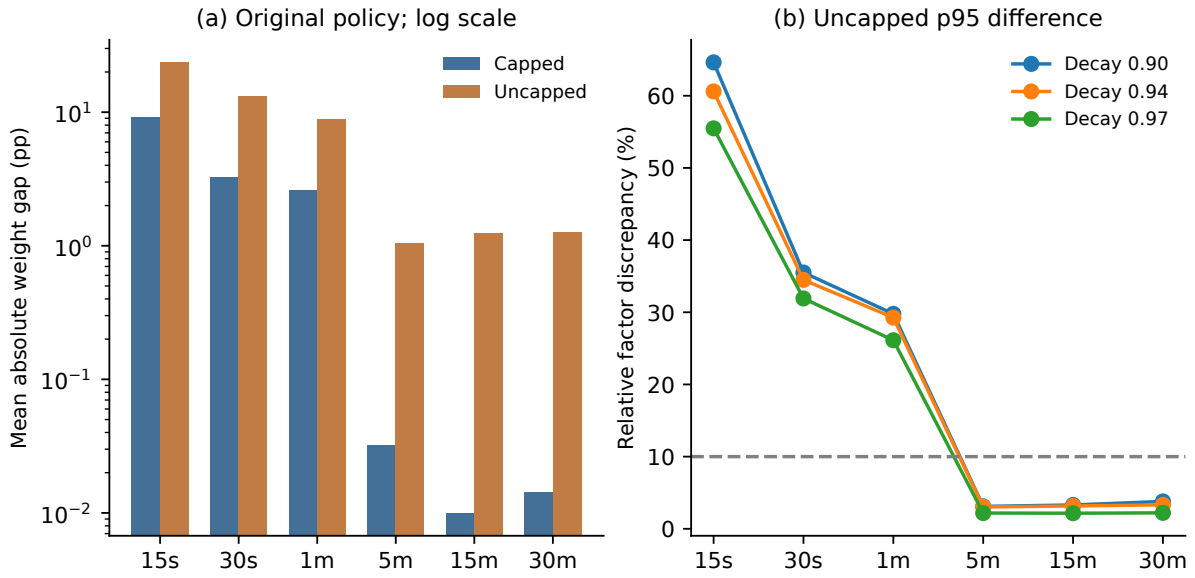


Figure 5: Cap masking and uncapped source differences. Panel (a) uses the original policy and a logarithmic vertical scale for mean absolute weight gaps. Panel (b) reports $100[\exp\{q_{.95}(|\log(u^L/u^M)|)\} - 1]$ at each decay. The sample percentile is descriptive, not an inferential interval. The target and annualization cancel in the uncapped ratio. Source: A11, explicitly post-result.

states, not measurements of true risk or future discrepancy probabilities.

Across all 27 fixed target/cap/decay settings, at least ten origins exceed a 5pp absolute gap in 18 one-minute policies and 2 five-minute policies. Some policies produce no constrained gap because both inputs saturate. Figure 6 and Appendix B report the entire specified grid, including such zeros; no policy is selected as optimal. The 27 policies share data and formulas and are not independent replications.

For additional context, the linear EWMA shock diagnostic attributes 19.41% of one-minute absolute propagation potential to the largest single date and 66.11% to the five largest dates at decay .94. These are concentration shares of an absolute triangle-bound measure. They are not shares of nonlinear capped weight losses, independent shocks or causal economic damage. The source-state difference itself allows signed cancellation.

6.1 Broader asset coverage qualifies the SPY result

The fixed A13 follow-up extends QQQ, IWM, EFA and TLT to the same 398-session calendar. It adds 1,508 source-paired fund-days and verifies/reuses the 84 January observations already seen in A10, yielding 1,592 ETF pairs. Together with SPY, the study now contains 1,990 distinct fund-days; the reused January panel is counted once. This design was frozen before the additional acquisition, after the SPY result was known. It is a follow-up exploratory breadth study, not untouched chronological validation.

Primary ETF incidence excludes the already observed January 2014 panel and both fixed discovery dates. This leaves 376 eligible sessions per fund before grid incompleteness. Full-calendar slopes and initialized EWMA histories retain January and the discovery dates to preserve spacing and state history. The same paired resampled date indices are shared across all funds, constructions and grids; assets are not treated as independent experiments. Appendix F reports every primary construction/grid count and its valid denominator. All three slope bands, all fixed-risk summaries

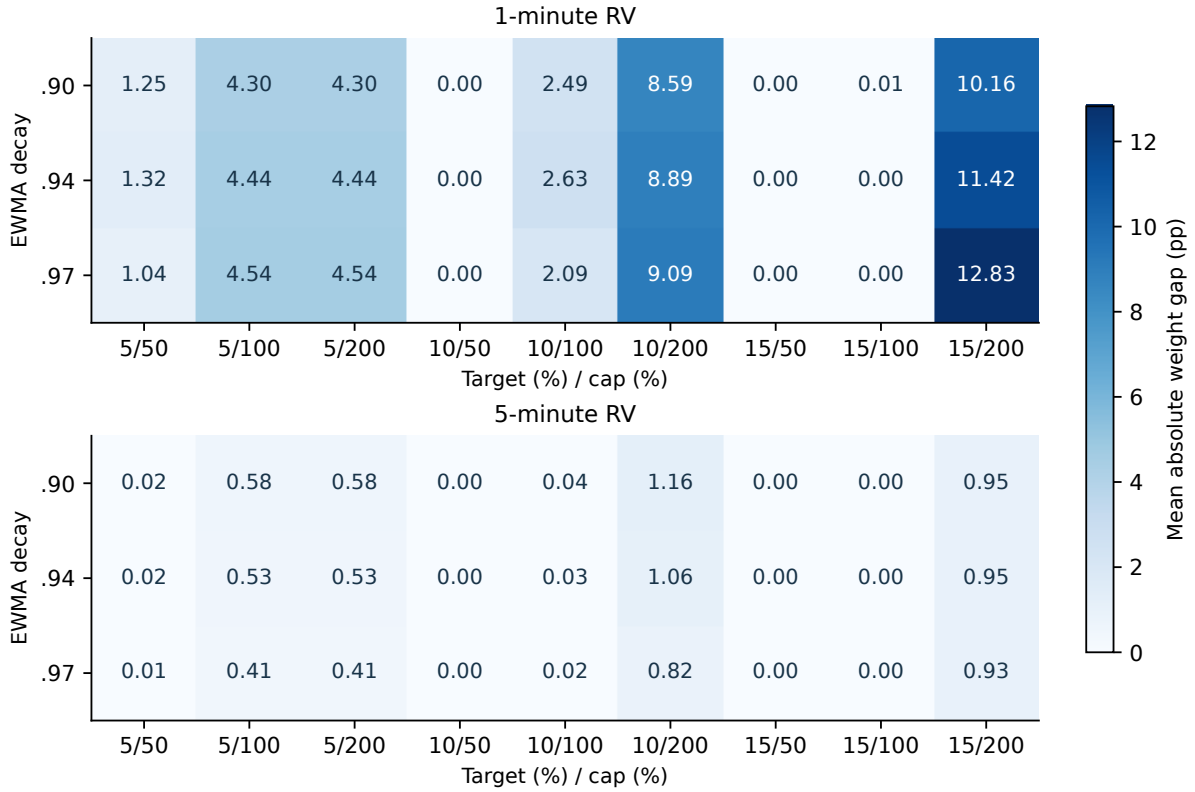


Figure 6: Every fixed policy setting at one and five minutes. Cells show mean absolute source weight gaps in pp. Both panels share a color scale. Targets/caps are scenario parameters; filled cells do not represent selected strategies. The specified post-result exercise changes no source filtering or observations. Source: A11.

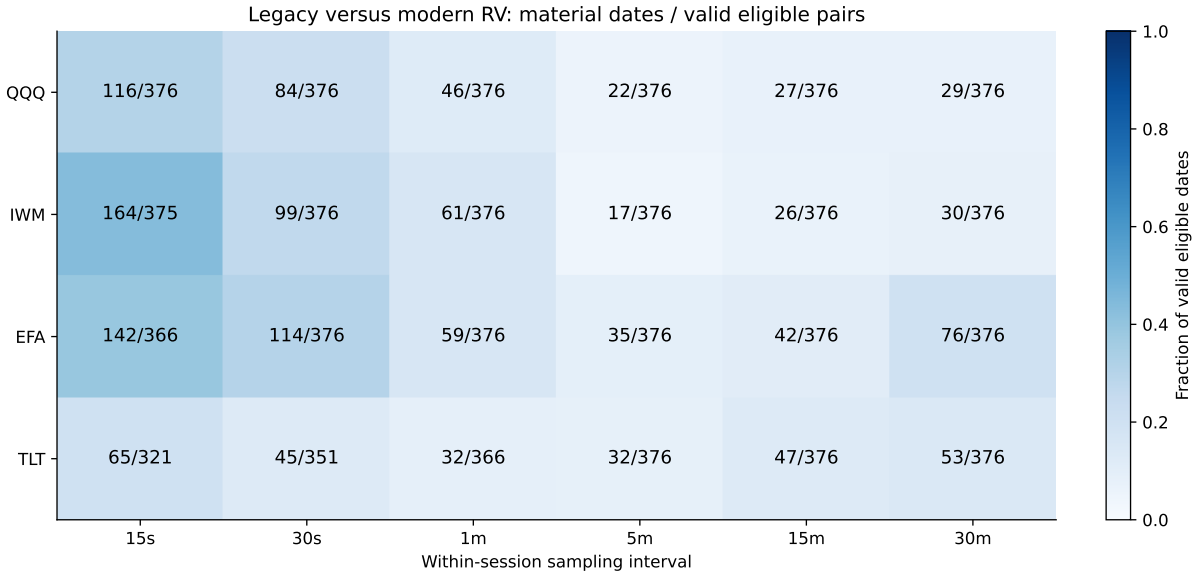


Figure 7: A13 legacy/modern RV incidence across every fixed fund and grid. Cells show material dates over valid eligible pairs; color is their fraction on a common zero-to-one scale. The threshold is the same factor 1.1 and its reciprocal. January 2014 and the explicit discovery dates are excluded. Missing pairs never become agreement; the reference is not verified true variance.

and all block-length intervals are supplied as machine-readable supplements.

Table 3 changes the practical interpretation. At five minutes, QQQ has 50 of 272 dependent origins above the fixed 5pp gap threshold, whereas the original SPY policy has 0. The QQQ mean capped gap is 1.8268pp, against 3.4166pp before the cap. Both funds use the same calendar, initialization, warmup and policy. The SPY finding therefore cannot support a general claim that five-minute sampling makes this source issue negligible. These comparisons concern proposed risk inputs, not realized returns or a superior trading strategy.

The opposite diagnostic problem appears in EFA. Its one-minute mean uncapped gap is 3.2885pp, but the capped gap is only 0.0109pp: 99.67% attenuation. A near-zero constrained gap can coexist with a nonzero underlying proposal gap. IWM shows a larger one-minute capped discrepancy and much less cap attenuation, while TLT's one-minute full-history risk calculation is withheld: 10 eligible legacy/reference pairs have incomplete grids. This is a data-quality limitation, not a zero policy effect. No missing dates are filled or removed to construct a complete EWMA history.

The crossed constructions also qualify a universal category-B explanation. For QQQ at one minute, B restoration retains 46 material RV dates, whereas modern condition information with legacy time leaves 4. The SPY restoration result thus does not identify a single category-level mechanism for every fund. The broader evidence favors auditing complete source information and its interactions, rather than applying an asset-independent repair inferred from SPY. Exact economic meanings of every mounted field remain a separate documentation question.

The extension passed 6,503 exported-member hash checks, independent calendar/coverage checks and daily correspondence/measurement checks. A separate result implementation recalculated 6,445 scalars, using explicit EWMA convolution and an independently constructed restart-segment sampler, and rejected a planted altered incidence. This verifies the reported calculations within the frozen design; it does not make the historical window a new regime or certify either source as economic truth.

Table 3: Every A13 fund at one and five minutes under the original fixed policy. RV/RQ entries are material/valid primary pairs. Weight gaps are mean absolute percentage-point differences; "large" counts dependent origins with gap at least 5pp. Nonmissing histories have 272 origins. The incomplete TLT one-minute history is withheld.

Fund	Grid	RV hits/valid	RQ hits/valid	Capped pp	Uncapped pp	Large origins
QQQ	1m	46/376	34/376	5.8531	7.5218	133
QQQ	5m	22/376	13/376	1.8268	3.4166	50
IWM	1m	61/376	48/376	7.7539	7.7539	186
IWM	5m	17/376	8/376	1.2386	1.2394	9
EFA	1m	59/376	19/376	0.0109	3.2885	0
EFA	5m	35/376	2/376	0.0101	1.0874	0
TLT	1m	32/366	18/366	withheld	withheld	withheld
TLT	5m	32/376	3/376	0.0157	0.8150	0

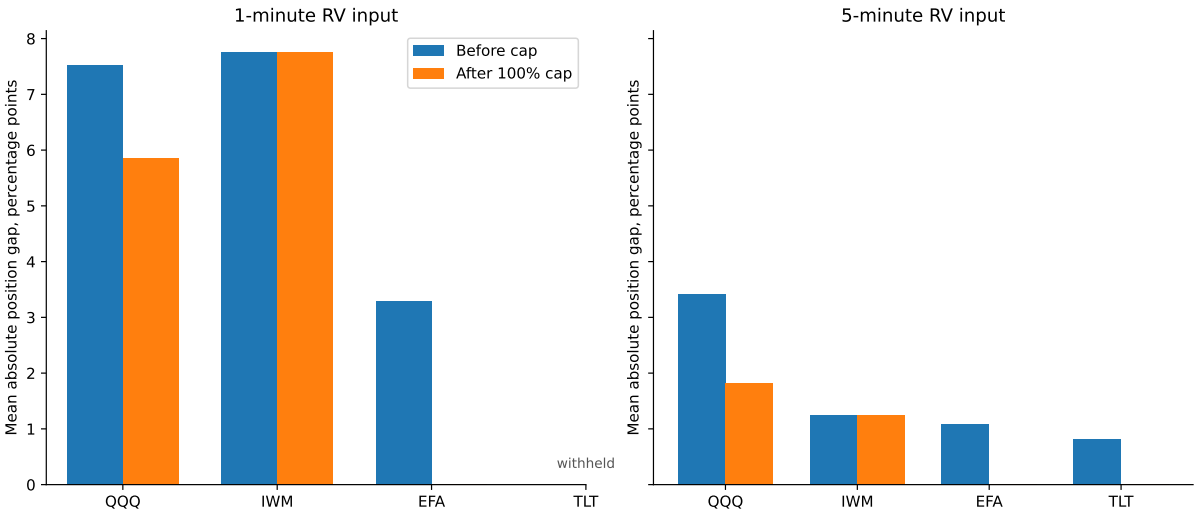


Figure 8: A13 fixed-policy gaps before and after the common cap, with shared vertical scale. Mean absolute gaps are in percentage points, not returns. Asset-specific variance levels affect how often the cap binds. A large attenuation percentage must be read together with its absolute uncapped gap. No policy is selected as optimal.

7 Discussion and limitations

The results distinguish three questions that can otherwise be conflated. First, do sources preserve the same information under a reused literal filter? Second, do the resulting measurement/state estimates agree? Third, do constrained decisions agree? Matching the last object cannot certify the first two. The combination of crossed source constructions and an explicit exposure-cap diagnostic is a practical audit workflow for empirical finance.

The study meets its predefined engineering relevance criterion, but that is a modest conclusion. It does not establish academic priority, a new estimator or a production-optimal cleaning rule. Existing work already establishes measurement-induced apparent roughness and historical TAQ compatibility problems. The value here is the dataset-specific controlled evidence, transparent source pipeline and independently checked propagation into a decision input.

Several limitations constrain generalization. The sample is historical and determined by the mounted common source overlap. Source regimes, timestamp precision, condition coverage and market structure may change. The expanded ETF panel shares the same historical source regime and cannot resolve these concerns. Stationary date-block intervals should therefore be read cautiously rather than as unconditional guarantees. The source reference is not verified integrated variance, and observed-row correspondence is weaker than unique execution identity.

The strict filter and opening convention are also consequential. The filter excludes conditions retained by broader cleaning defaults, while the dominant-venue and simultaneous-price choices interact with the source representation. The B-restoration exercise identifies a narrow mechanism only within this design. No silent deletion, threshold adjustment or guessed quarticity rescaling is used to create agreement. A future cleaning-rule comparison would be a separately specified study, not an interpretation retrofitted to this one.

Finally, the risk comparison is intentionally not an economic backtest. It uses fixed states and scenario weights, no fitted model-selection contest, and no returns, costs, cash, overnight or execution accounting. The smaller five-minute SPY discrepancies do not generalize uniformly across the extended funds and do not establish the best sampling interval. The original registered 1,250-day forecast comparison remains deferred for source/history feasibility reasons and is not presented as completed by this case study. An option implied-volatility surface is not required to answer the paper's realized-measurement question.

8 Verification, reproducibility and disclosure

Verification is layered rather than asserted universally. The source export checks file identity, fixed sample coverage and per-construction measurement-check records. The moment pipeline compares SQL aggregation with an independent NumPy construction, including expected/observed grid counts. An independent calendar check can reject a shared early-close error that source agreement alone would miss. Matching ordered records is checked separately from numerical RV/RQ agreement.

At the result level, independent standard-library arithmetic recomputes 900 A10 scalars. The A11 check uses original paired inputs, direct EWMA convolution and explicit shock propagation rather than the analysis recurrence and closed-form influence masses. It checks 10,584 scalars and 96 overlapping original-A10 values, and rejects a planted altered policy output. Synthetic tests cover selected covariance, scaling, missingness, initialization, causality and boundary cases. Appendix A maps the checks to components and their limits. These checks support the recorded calculations; they do not establish economic truth, undocumented vendor semantics or the absence of every bug.

The supporting derivations have separate checks using exact Gaussian polynomial quadrature,

Gamma-function moments, numerical differentiation of log variograms, initialized convolution and cap boundary cases. These verify selected identities under their stated assumptions; they do not verify that the assumptions hold for the empirical data.

The paper's numerical macros and tables are generated from recorded outputs and have a separate machine-readable claim map. A paper-specific independent checker verifies their source selectors, transformations and printed values. The source code, input/output hashes and run IDs support a concrete restart path. A separate synthetic-only demo/candidate runs without WRDS access or vendor records. Restricted daily data and this private empirical working history are not automatically public-release material.

Data availability. The empirical inputs are licensed historical WRDS source data. Raw ticks and vendor observations are not redistributed. The accompanying repository supplies processing source, synthetic examples and the manuscript. Reproducing the empirical findings requires separately licensed inputs, which the repository does not distribute. The repository omits private research history and source observations. Closed validation/final observations have not been used.

AI and authorship disclosure. Martin Valachy set the research direction, scope and project goals. Codex implemented the computational pipeline, automated independent numerical checks, figures and this draft. Agreement between separately implemented checks is computational verification, not independent human peer review. Martin's comprehension/code review is not asserted complete. The working-paper form is intended to support that review and an honest undergraduate portfolio, not to imply journal acceptance or work performed without AI assistance.

9 Conclusion

A source-compatible pipeline must preserve information, not merely reuse the same literal filter. In this fixed historical comparison, time/condition representations materially change realized measures and descriptive roughness statistics, and those differences propagate into a persistent one-minute risk-input episode. A common exposure cap can then hide disagreement between the underlying state estimates. The uncapped SPY check qualifies its smaller five-minute effects; the fixed ETF follow-up further shows that this sampling-frequency reassurance is asset-dependent. QQQ retains sizable five-minute proposed-weight gaps, while EFA illustrates strong one-minute cap masking and incomplete TLT grids remain withheld.

For applied work, the defensible lesson is to audit source fields, session counts, measurement statistics, uncapped states and constrained inputs as separate objects. The evidence supports that practical workflow. It does not select a universally best estimator, identify latent market roughness or demonstrate a profitable strategy.

References

[1] Gatheral, J., Jaisson, T., and Rosenbaum, M. (2014). *Volatility is rough*. arXiv:1410.3394, version 1. https://arxiv.org/abs/1410.3394.

[2] Cont, R., and Das, P. (2023). *Rough volatility: fact or artefact?* arXiv:2203.13820, version 3. https://arxiv.org/abs/2203.13820v3.

[3] Fukasawa, M., Takabatake, T., and Westphal, R. (2019). *Is Volatility Rough?* arXiv:1905.04852, version 2. https://arxiv.org/abs/1905.04852v2.

[4] Holden, C. W., and Jacobsen, S. (2014). Liquidity Measurement Problems in Fast, Competitive Markets: Expensive and Cheap Solutions. *The Journal of Finance*, 69(4), 1747–1785. https://doi.org/10.1111/jofi.12127.

[5] Cipollini, F., Cruciani, G., Gallo, G. M., Insana, A., Otranto, E., and Spagnolo, F. (2026). *VOLatility Archive for Realized Estimates (VOLARE)*. arXiv:2602.19732, version 1. https://arxiv.org/html/2602.19732v1.

[6] NYSE Euronext (2014). *Daily TAQ Client Specification*, version 2.0, 28 July 2014, trade-condition fields, pp. 26–29 of the PDF. https://people.stern.nyu.edu/jhasbrou/Teaching/POST%202015%20Spring/TAQ/Daily_TAQ_Client_Spec_v2%200.pdf.

[7] `highfrequency` maintainers (accessed 22 October 2026). *dataHandling.R*: trade-condition cleaning and related data-handling source. https://github.com/jonathancornelissen/highfrequency/blob/master/R/dataHandling.R. The inspected branch is mutable; the project records the inspected source location and date.

[8] Politis, D. N., and Romano, J. P. (1994). The Stationary Bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313. https://doi.org/10.1080/01621459.1994.10476870.

[9] Bennedsen, M., Lunde, A., and Pakkanen, M. S. (2022). Decoupling the Short- and Long-Term Behavior of Stochastic Volatility. *Journal of Financial Econometrics*, 20(5), 961–1006. https://doi.org/10.1093/jjfinec/nbaa049. Inspected author version: https://arxiv.org/abs/1610.00332v3.

[10] Barndorff-Nielsen, O. E., and Shephard, N. (2002). Estimating quadratic variation using realized variance. *Journal of Applied Econometrics*, 17(5), 457–477. https://doi.org/10.1002/jae.691. Inspected author copy, Sections 1 and 3: https://shephard.scholars.harvard.edu/sites/g/files/omnuum7741/files/jae02.pdf.

A Component verification and run map

Table 4: What was checked, and what remains outside that check.

Component	Evidence	Limit
Source/sample	Fixed calendar/panel, exported member hashes, per-file source alignment flags	Identity is observable-row correspondence, not unique execution identity
RV/RQ construction	SQL versus NumPy; exact grid counts; missingness; synthetic unit/early-close errors	Does not certify a universally best trade filter or true integrated variance
Finite-band slopes	Independent log-variogram/OLS arithmetic; scale invariance; calendar holes	No local market-H identification or empirical H coverage claim
EWMA and weights	Recurrence versus direct convolution; causality, initialization and fixed-policy overlaps	No return, cash, overnight, cost or execution evaluation
Policy/influence	Every fixed setting; explicit shock sums; altered summary rejected	Post-result; no policy-selection or nonlinear causal-loss attribution
Synthetic baseline	Known covariance/measurement checks; independent table arithmetic; coupled refinement	Conditional simulator evidence, not empirical model validation
Paper/presentation	Source-linked numeric facts, compilation, references and visual inspection	Cannot substitute for human comprehension and research review

The primary provenance anchors are:

A08 simulation `side_project_study_20261021T214152Z`.

A08 independent tables `side_project_table_check_20261021T215134Z`.

A08 refinement `side_project_refinement_20261021T214206Z`.

A10 full analysis `source_replication_analysis_20261022T014736Z`.

A10 independent results `source_result_check_20261022T015029Z`.

A10 export provenance `source_export_verified_20261022T014656Z`.

A11 sensitivity `source_policy_robustness_20261022T032458Z`.

A11 independent results `source_policy_check_20261022T032503Z`.

Derivation checks `paper_derivation_check_20261022T121236Z`.

The source configurations and methods are recorded in `configs/side_project.json`, `configs/source_replication.json`, `configs/source_policy_robustness.json` and the associated A08/A10/A11 protocol files. A11 was frozen locally before computing its sensitivities, after the original A10 outcome was known. The exploratory chronology is retained rather than relabeled as a confirmatory design.

B All fixed one- and five-minute policies

Table 5: Complete A11 policy matrix. Targets/caps are percentages; gaps are mean absolute weight differences in pp; "large" counts origins with an absolute gap of at least 5pp. All rows use the same 272 dependent origins.

λ	Target %	Cap %	1m gap (pp)	1m large	5m gap (pp)	5m large
0.90	5	50	1.247	26	0.021	0
0.90	5	100	4.296	92	0.578	1
0.90	5	200	4.296	92	0.578	1
0.90	10	50	0.000	0	0.000	0
0.90	10	100	2.494	37	0.042	0
0.90	10	200	8.592	130	1.157	9
0.90	15	50	0.000	0	0.000	0
0.90	15	100	0.010	0	0.000	0
0.90	15	200	10.162	116	0.954	11
0.94	5	50	1.316	29	0.016	0
0.94	5	100	4.443	86	0.528	0
0.94	5	200	4.443	86	0.528	0
0.94	10	50	0.000	0	0.000	0
0.94	10	100	2.632	47	0.032	0
0.94	10	200	8.885	147	1.055	7
0.94	15	50	0.000	0	0.000	0
0.94	15	100	0.000	0	0.000	0
0.94	15	200	11.422	142	0.949	13
0.97	5	50	1.044	19	0.011	0
0.97	5	100	4.543	77	0.409	0
0.97	5	200	4.543	77	0.409	0
0.97	10	50	0.000	0	0.000	0
0.97	10	100	2.088	55	0.023	0
0.97	10	200	9.087	183	0.818	0
0.97	15	50	0.000	0	0.000	0
0.97	15	100	0.000	0	0.000	0
0.97	15	200	12.826	173	0.931	1

C All fixed slope bands

Table 6: Full-window finite-band H-tilde at every grid and fixed band. Both explicit discovery dates remain included. These values are descriptive and do not carry latent-H confidence intervals.

Largest lag	Grid	Modern	Legacy	B restored
5	15s	0.1921	0.0913	0.1800
5	30s	0.1866	0.1532	0.1839
5	1m	0.1773	0.1546	0.1763
5	5m	0.1742	0.1708	0.1709
5	15m	0.1538	0.1492	0.1518
5	30m	0.1389	0.1370	0.1386
10	15s	0.1703	0.0828	0.1644
10	30s	0.1629	0.1228	0.1624
10	1m	0.1542	0.1239	0.1516
10	5m	0.1409	0.1409	0.1391
10	15m	0.1046	0.1014	0.1019
10	30m	0.0936	0.0912	0.0916
22	15s	0.1438	0.0766	0.1397
22	30s	0.1342	0.1027	0.1331
22	1m	0.1253	0.1046	0.1228
22	5m	0.1021	0.1039	0.1014
22	15m	0.0802	0.0772	0.0783
22	30m	0.0643	0.0618	0.0627

D Derivations and interpretation

This appendix supplies the mathematical bridge from a sampled price to a reported decision input. The arguments below are conditional calculations under stated assumptions, not fitted models of the historical sample. They explain what the diagnostics can reveal and why the empirical comparisons do not identify latent market volatility.

D.1 From price increments to realized variance and quarticity

Use a trading-session clock $u \in [0, 1]$, with the actual session rescaled to unit length. In an ideal continuous-price model,

$$dp_u = b_u\, du + \sigma_u\, dW_u, \qquad IV = \int_0^1 \sigma_u^2\, du, \qquad IQ = \int_0^1 \sigma_u^4\, du. \tag{5}$$

Here p is log price, IV is integrated variance and IQ is integrated quarticity. Quadratic-variation theory motivates sums of squared returns; the continuous stochastic-volatility setting supplies additional distributional results. Barndorff-Nielsen and Shephard [10] distinguish this consistency statement from assumptions needed to quantify precision (their Sections 1 and 3). The following finite-grid calculation makes the moment normalizations explicit.

For this calculation, ignore drift and suppose the variance path is deterministic, or independent of the Brownian price shocks. On a grid of M intervals, write

$$r_i = \sqrt{a_i} Z_i, \quad a_i = \int_{(i-1)/M}^{i/M} \sigma_u^2\, du, \quad Z_i \overset{\text{iid}}{\sim} N(0, 1). \tag{6}$$

The assumptions imply independent conditional increments. They are stronger than a general model with leverage; conditional Gaussianity is not asserted for arbitrary stochastic volatility. Since $\mathbb{E}Z_i^2 = 1$ and $\mathbb{E}Z_i^4 = 3$,

$$\mathbb{E}[\mathrm{RV} \mid a] = \sum_i a_i = IV, \tag{7}$$

$$\mathrm{Var}(\mathrm{RV} \mid a) = 2\sum_i a_i^2, \tag{8}$$

$$\mathbb{E}[\mathrm{RQ} \mid a] = M\sum_i a_i^2, \qquad \mathrm{RQ} = \frac{M}{3}\sum_i r_i^4. \tag{9}$$

The variance uses $\mathrm{Var}(Z_i^2) = 3 - 1 = 2$ and vanishing cross covariances. For a sufficiently regular variance path, the last expectation is a Riemann approximation to IQ. With constant variance $a_i = IV/M$, it is exactly IV^2. This derives the factor $M/3$ rather than treating it as a vendor convention. Under the stated finite-grid model, $2\,\mathrm{RQ}/M$ is an unbiased estimator of the conditional variance in Equation (8).

Quarticity responds particularly strongly to isolated increments: multiplying one return by k multiplies its RV contribution by k^2 and its RQ contribution by k^4. That does not mean that a larger source RQ discrepancy always occurs: retained trades, signs, neighboring returns and grid endpoints all matter. The complete incidence table is therefore necessary.

The session clock also explains why M must follow the actual session. If a complete shortened session has m returns but its quarticity is incorrectly multiplied by a full-session count M, the computed RQ is M/m times the intended session-normalized RQ, even when the prices and returns are identical. Missing intervals cannot be repaired by substituting the observed count into the formula: the resulting returns may cover an irregular or truncated interval. This is why incomplete grids are withheld rather than silently renormalized.

For decimal log returns, RV has squared-return units and RQ has fourth-power units. Converting all returns to percentage units multiplies RV by 100^2 and RQ by 100^4; changing the currency denomination of every price leaves log returns unchanged. Multiplying a daily RV by an annualization factor A is another explicit transformation. A daily RQ can be multiplied by A^2 to express the corresponding fourth-power scale; it does not then become an estimate of the integral of quarticity over an observed year. Units and observation windows must both agree before reconciliation.

D.2 What a relative measurement diagnostic does and does not estimate

The preceding conditional-variance identity motivates the dimensionless diagnostic

$$\widehat{q} = \frac{2\,\mathrm{RQ}}{M\,\mathrm{RV}^2} = \frac{2}{3}\frac{\sum_i r_i^4}{(\sum_i r_i^2)^2}. \tag{10}$$

A first-order delta approximation, $\log(\mathrm{RV}) - \log(IV) \simeq (\mathrm{RV} - IV)/IV$, connects a relative variance to log measurement error when that error is small. It does not establish iid additive errors across days, eliminate leverage, handle market microstructure noise or certify that $\widehat{q}$ itself is an unbiased log-error variance.

Even the constant-variance Gaussian case illustrates the distinction between a ratio of expectations and an expectation of a ratio. Let $P_i = Z_i^2/\sum_j Z_j^2$. Spherical symmetry, or the Dirichlet law with all parameters $1/2$, gives

$$\mathbb{E}[P_i^2] = \frac{(1/2)(3/2)}{(M/2)(M/2+1)} = \frac{3}{M(M+2)}, \qquad \mathbb{E}[\widehat{q}] = \frac{2}{M+2}. \tag{11}$$

In contrast, $\mathrm{Var}(\mathrm{RV} \mid IV)/IV^2 = 2/M$. The finite-sample discrepancy is a consequence of the random denominator. This diagnostic is supporting methodology, not an extra empirical endpoint of A10/A11. In particular, no empirical variogram is "corrected" by subtracting it.

D.3 Two different places where observation noise enters

First consider price-level noise on a fixed grid: $y_i = p_i + \eta_i$, where independent centered errors have variance ω^2 and are independent of the efficient price. Observed returns are $\widetilde{r}_i = r_i + \eta_i - \eta_{i-1}$. Expansion gives

$$\mathbb{E}[\widetilde{\mathrm{RV}} \mid a] = IV + 2M\omega^2. \tag{12}$$

The cross term has zero expectation and each differenced error has variance $2\omega^2$. Adjacent error differences are negatively correlated; independence across observed returns is unnecessary for this expectation. If the level noise is also Gaussian, each marginal observed return is Gaussian with variance $a_i + 2\omega^2$, hence

$$\mathbb{E}[\widetilde{\mathrm{RQ}} \mid a] = M\sum_i (a_i + 2\omega^2)^2 = M\sum_i a_i^2 + 4M\omega^2 IV + 4M^2\omega^4. \tag{13}$$

Higher sampling frequency can therefore increase contamination even while it improves the ideal noise-free discretization. This illustrative model is implemented in the supporting simulation; it is not a claim that the historical source discrepancy consists of independent Gaussian quote errors. In particular, source-specific condition information is structured, not an iid perturbation.

The second model concerns a daily log-variance proxy: $Y_t = X_t + e_t$. If e_t is centered, iid, independent of X, and has variance q, distinct endpoints give

$$V_Y(k) := \mathbb{E}[(Y_{t+k} - Y_t)^2] = V_X(k) + 2q. \tag{14}$$

To derive this, expand the square, drop the cross term by independence, and use $\mathrm{Var}(e_{t+k} - e_t) = 2q$. For stationary but correlated independent measurement error, the addition instead equals $2[\gamma_e(0) - \gamma_e(k)]$. If the error is related to X, an additional cross term remains. Price-level noise and daily log-proxy noise therefore cannot be identified with each other merely because both can flatten a plot.

D.4 Why the finite-band statistic is not latent roughness

For a positive population variogram, define a local scale diagnostic

$$H_{\mathrm{eff}}(\tau) = \frac{1}{2}\frac{d\log V(\tau)}{d\log \tau} = \frac{\tau V'(\tau)}{2V(\tau)}. \tag{15}$$

If $V(\tau) = C\tau^{2H}$ exactly, this equals H at every scale. Adding the iid daily log-noise floor produces

$$H_{\mathrm{eff},Y}(\tau) = H\frac{C\tau^{2H}}{C\tau^{2H} + 2q}. \tag{16}$$

For positive q, the factor lies between zero and one. This is a precise attenuation mechanism under this model, not a proof that every empirical low slope is noise.

For a stationary OU log-variance process with covariance $v\exp(-\kappa|\tau|)$, stationarity gives $V_X(\tau) = 2v[1 - \exp(-\kappa\tau)]$. Differentiating yields

$$H_{\mathrm{eff},X}(\tau) = \frac{\kappa\tau}{2(\exp(\kappa\tau) - 1)}, \qquad H_{\mathrm{eff},Y}(\tau) = \frac{\tau v\kappa\exp(-\kappa\tau)}{2v[1 - \exp(-\kappa\tau)] + 2q}. \tag{17}$$

Since $\exp(z) - 1 \sim z$ near zero, the noise-free local limit is 1/2. It falls at longer scales because OU increments approach their stationary variance. An OU process is locally diffusive, not differentiable. "Nonrough" here distinguishes it from a volatility model with a smaller local exponent; it does not mean a smooth deterministic path. A finite nonzero noise floor further changes the small-scale behavior.

The reported estimator uses finitely many calendar lags, not a derivative at zero. With $x_k = \log k$, $z_k = \log \widehat{V}(k)$, its explicit formula is

$$\widetilde{H}_K = \frac{1}{2} \frac{\sum_{k=1}^{K} (x_k - \bar{x})(z_k - \bar{z})}{\sum_{k=1}^{K} (x_k - \bar{x})^2}. \tag{18}$$

It recovers H for an exact power law on that band, but otherwise summarizes the curvature and errors at the selected lags. Lagged increments overlap, their estimates are dependent, and different lags can have different valid endpoint counts. Ordinary regression standard errors would not automatically be valid. Missing calendar dates are not compressed, because doing so changes the meaning of k. The known-truth experiment isolates this distinction without fitting a market model.

Multiplying every daily RV by the same positive constant adds a constant to every log RV and leaves all increments and $\widetilde{H}_K$ unchanged. A source discrepancy that changes the slope must therefore have more structure than a constant variance-unit multiplier. This invariance does not identify which source is economically preferable.

D.5 Controlled source contrasts and interactions

Let $F(T, C)$ be any positive daily measurement produced by the entire fixed pipeline from time representation T and condition representation C. The crossed construction changes inputs to this function while retaining the observed price/size/venue sequence. On log scale, define

$$D_T = \log F(L, M) - \log F(M, M), \tag{19}$$

$$D_C = \log F(M, L) - \log F(M, M), \tag{20}$$

$$I = \log F(L, L) - \log F(L, M) - \log F(M, L) + \log F(M, M). \tag{21}$$

Then the legacy/reference log difference equals $D_T + D_C + I$ by algebra. An additive representation would set $I = 0$, but the pipeline provides no reason to assume this. Changing conditions can change retained-venue counts, the selected venue, same-time medians and previous-tick endpoints. Moreover, the materiality indicator thresholds an absolute difference, so incidence counts cannot be added as if they were signed effects. The paper reports the complete crossed constructions rather than claiming an additive causal decomposition. B restoration is a further controlled diagnostic within this function, not a universally justified trade deletion.

D.6 State propagation, initialization and persistence

Starting from $s_0 = \mathrm{RV}_0$, repeated substitution into Equation (3) gives

$$s_t = \lambda^t \,\mathrm{RV}_0 + (1 - \lambda) \sum_{i=1}^{t} \lambda^{t-i} \,\mathrm{RV}_i \,. \tag{22}$$

The coefficients sum to one. The initial coefficient is λ^t, not $(1 - \lambda)\lambda^t$; treating initialization like a later shock would lose mass. This direct convolution is the independent check on the recursive implementation.

For source differences $\delta_i = \mathrm{RV}_i^L - \mathrm{RV}_i^M$, subtraction gives exactly the same linear filter of δ_i. If only a later observation $j > 0$ differs, its effect at $t \geq j$ is $(1 - \lambda)\lambda^{t-j}\delta_j$. Its half-life, in sessions,

is $\log(1/2)/\log\lambda$. One discrepant input can therefore create many successive differing states; counting affected origins does not count independent source shocks. The paper labels weights by their information origin: s_t only supplies the session-$t+1$ input, avoiding use of the same day's complete RV before it exists.

For evaluated origins $\mathcal{T}$, put $a_{t,0} = \lambda^t$ and $a_{t,i} = (1-\lambda)\lambda^{t-i}$ for $1 \le i \le t$. The triangle inequality yields

$$\sum_{t\in\mathcal{T}} |s_t^L - s_t^M| \le \sum_i |\delta_i| \sum_{\substack{t\in\mathcal{T}\\ t\ge i}} a_{t,i}. \tag{23}$$

The right-hand summands define the absolute propagation-potential shares. They omit signed cancellation and precede the nonlinear state-to-weight transformation. Their concentration is informative about sensitivity, but is not an allocation of actual portfolio losses.

D.7 From variance states to weights and cap masking

Let $s_t > 0$ be a daily variance state, A a common annualization factor, $v_* > 0$ an annual volatility target and $c > 0$ a cap. Equating an unconstrained position's estimated annual volatility to its target gives

$$u_t\sqrt{As_t} = v_*, \qquad u_t = \frac{v_*}{\sqrt{As_t}}, \qquad \frac{\partial \log u_t}{\partial \log s_t} = -\frac{1}{2}. \tag{24}$$

The sensitivity follows by differentiating $\log u_t = \log v_* - (\log A + \log s_t)/2$. A small relative variance change therefore produces approximately half as large an oppositely signed relative unconstrained weight change. The exact ratio is

$$\frac{u_t^L}{u_t^M} = \sqrt{\frac{s_t^M}{s_t^L}}, \qquad \left|\log\frac{u_t^L}{u_t^M}\right| = \frac{1}{2}\left|\log\frac{s_t^L}{s_t^M}\right|. \tag{25}$$

Common targets and annualization cancel. If an entire source RV path is multiplied by $a > 0$, consistent initialization makes its state exactly a times as large and its unconstrained weight $a^{-1/2}$ times as large. Thus a constant unit error leaves the roughness slope unchanged while changing the risk input, illustrating why both audits are needed.

Now set $w_t = \min(c, u_t)$. The states at which a weight saturates satisfy

$$s_t \le \frac{v_*^2}{Ac^2}. \tag{26}$$

Within this region the constrained weight is locally insensitive to the state. If both sources are there, their weights agree exactly regardless of their state ratio. If neither is there, capping changes neither weight. If just one is there, the cap reduces the gap. These three exhaustive cases prove

$$|\min(c,a) - \min(c,b)| \le |a-b| \qquad (a,b>0). \tag{27}$$

The aggregate attenuation is defined as one minus the mean capped absolute gap divided by the mean uncapped absolute gap, when the latter is positive. Its value depends on the chosen target and cap. This makes the uncapped state-ratio diagnostic necessary for interpreting apparent decision agreement. It does not turn either estimated state into true risk or establish an optimal portfolio policy.

D.8 What the block intervals quantify

For a source pair and grid, let J_t indicate a valid comparison and I_t indicate both validity and a discrepancy above the fixed threshold. The empirical incidence is $\widehat{p} = \sum_t I_t / \sum_t J_t$, with the two

discovery dates ineligible for both sums. Resampling the numerator and denominator on the same calendar indices preserves this definition; resampling only discrepancies or compressing to valid dates changes the experiment.

The stationary bootstrap [8] starts at a uniformly drawn calendar index. Each subsequent index restarts uniformly with probability $1/L$, or continues to the next circular calendar index otherwise. Thus $\Pr(B = b) = (1 - 1/L)^{b-1}/L$ for block length B, with mean L. Using identical indices across paired objects retains their contemporaneous dependence. Recomputing the incidence on each resample yields the displayed percentile intervals; block-length sensitivities assess one aspect of dependence handling.

These intervals rely on a useful stationary, weak-dependence approximation and the representativeness of the available window. They do not supply new independent episodes, remove a source-regime change, establish source equivalence or account for every exploratory choice. The slope panels are descriptive and are not assigned these incidence intervals. The policy matrix is a systematic post-result sensitivity, not a family of successful independent hypothesis tests.

E Reproduction and boundaries

The private original A10 export is kept outside the repository. From the authorized research workspace, the empirical pipeline is checked using `verify_source_export.py`, `analyze_source_replication.py`, `check_source_results.py`, `analyze_policy_robustness.py` and `check_policy_robustness.py`. Each receives the private-folder argument. The paper preparation script reads these checked outputs, writes named facts/tables and verifies the displayed weight paths against the original summary. A separate checker reads the source records using standard-library arithmetic and compares the generated fact macros.

The PDF is compiled from editable LaTeX sources. The delivery includes a claim map, bibliography and verification summary; the wider local atlas is supplementary and not a substitute for the paper's argument. The portable synthetic demo has its own isolated installation procedure. Reproduction generates new run IDs and timing metadata; scientific tolerances, not byte equality across numerical platforms, determine arithmetic agreement. No older mixed-period audit script is part of these instructions.

A13 is reproduced from its separate licensed aggregate export using `verify_etf_breadth.py`, `analyze_etf_breadth.py` and `check_etf_breadth.py`. The exported-member fingerprint, fixed calendar and previously seen January hashes are checked before analysis. `prepare_breadth_paper.py` reads only its checked summary tables; `check_breadth_paper.py` independently checks every additional numerical macro and catches a planted altered fact. The complete summary tables accompany the manuscript without the underlying daily observations.

The original broader forecast contest and unresolved source-reconciliation gates are preserved as deferred historical plans. No shortened-window forecast winner is reported. The option-surface extension was parked after clarification of the paper's scope, and no option observations were acquired. No final-period, publication, spending or contact action is required by this reproduction path.

F Complete A13 primary counts and slope context

This appendix retains every fund, grid and construction. "Time only" means modern time with legacy flags; "Cond. only" means legacy time with modern flags. B restoration is a diagnostic, not a production rule. Every incidence entry is material/valid eligible pairs. The primary eligible calendar has 376 dates per fund; valid denominators can be smaller. January 2014 is previously observed and is excluded here, together with the fixed discovery dates. Missingness is not classified as agreement.

The frozen design is `protocol/A13_ETF_BREADTH.md`, with configuration `configs/etf_breadth.json`. Its provenance anchors are:
Export check `etf_breadth_export_verified_20261022T144522Z`.
Analysis `etf_breadth_analysis_20261022T144854Z`.
Independent results `etf_breadth_independent_check_20261022T145028Z`.

Table 7: Complete A13 RV primary incidence. Materiality is an absolute log ratio above log(1.1), equivalently a ratio outside $[1/1.1, 1.1]$. Modern/reference comparisons have zero material dates but can have missing grids.

Fund	Grid	Modern	Legacy	Time only	Cond. only	B restored
QQQ	15s	0/376	116/376	125/376	3/376	116/376
QQQ	30s	0/376	84/376	81/376	5/376	84/376
QQQ	1m	0/376	46/376	46/376	4/376	46/376
QQQ	5m	0/376	22/376	15/376	7/376	22/376
QQQ	15m	0/376	27/376	5/376	19/376	27/376
QQQ	30m	0/376	29/376	6/376	22/376	29/376
IWM	15s	0/375	164/375	171/375	1/375	14/375
IWM	30s	0/376	99/376	103/376	3/376	11/376
IWM	1m	0/376	61/376	57/376	2/376	6/376
IWM	5m	0/376	17/376	12/376	5/376	6/376
IWM	15m	0/376	26/376	9/376	16/376	21/376
IWM	30m	0/376	30/376	8/376	24/376	26/376
EFA	15s	0/366	142/366	164/366	3/366	110/366
EFA	30s	0/376	114/376	124/376	3/376	70/376
EFA	1m	0/376	59/376	55/376	8/376	36/376
EFA	5m	0/376	35/376	21/376	14/376	30/376
EFA	15m	0/376	42/376	18/376	24/376	37/376
EFA	30m	0/376	76/376	30/376	50/376	69/376
TLT	15s	0/321	65/321	73/321	5/321	34/321
TLT	30s	0/351	45/351	48/351	9/351	24/351
TLT	1m	0/366	32/366	23/366	12/366	23/366
TLT	5m	0/376	32/376	13/376	18/376	28/376
TLT	15m	0/376	47/376	16/376	26/376	40/376
TLT	30m	0/376	53/376	19/376	34/376	49/376

Table 8: Complete A13 RQ primary incidence. Materiality is an absolute log ratio above log(2), with reciprocal lower boundary. The larger threshold and fourth-power units differ from RV; these counts do not estimate a latent-noise distribution.

Fund	Grid	Modern	Legacy	Time only	Cond. only	B restored
QQQ	15s	0/376	113/376	114/376	4/376	113/376
QQQ	30s	0/376	75/376	68/376	4/376	75/376
QQQ	1m	0/376	34/376	34/376	1/376	34/376
QQQ	5m	0/376	13/376	11/376	2/376	13/376
QQQ	15m	0/376	6/376	2/376	2/376	6/376
QQQ	30m	0/376	2/376	0/376	2/376	2/376
IWM	15s	0/375	157/375	151/375	0/375	10/375
IWM	30s	0/376	81/376	89/376	0/376	7/376
IWM	1m	0/376	48/376	51/376	1/376	4/376
IWM	5m	0/376	8/376	10/376	1/376	1/376
IWM	15m	0/376	2/376	4/376	0/376	0/376
IWM	30m	0/376	0/376	1/376	0/376	0/376
EFA	15s	0/366	53/366	61/366	3/366	12/366
EFA	30s	0/376	39/376	37/376	5/376	12/376
EFA	1m	0/376	19/376	17/376	3/376	7/376
EFA	5m	0/376	2/376	1/376	1/376	2/376
EFA	15m	0/376	4/376	2/376	2/376	3/376
EFA	30m	0/376	7/376	2/376	5/376	6/376
TLT	15s	0/321	26/321	17/321	10/321	16/321
TLT	30s	0/351	24/351	15/351	10/351	18/351
TLT	1m	0/366	18/366	10/366	10/366	15/366
TLT	5m	0/376	3/376	1/376	4/376	4/376
TLT	15m	0/376	8/376	2/376	7/376	7/376
TLT	30m	0/376	11/376	1/376	7/376	9/376

Table 9: A13 descriptive H-tilde over lags 1–10 for all fixed constructions/grids. Full 398-session calendar including previously seen January/discovery dates. Missing values retain calendar endpoints, so comparisons may have different endpoint support; the machine-readable H table also gives common-endpoint contrasts and all lag counts. No latent-H confidence interval is implied.

Fund	Grid	Modern	Legacy	Time only	Cond. only	B restored
QQQ	15s	0.1744	0.0668	0.0630	0.1765	0.0668
QQQ	30s	0.1670	0.0795	0.0648	0.1708	0.0795
QQQ	1m	0.1576	0.0931	0.0772	0.1604	0.0931
QQQ	5m	0.1275	0.1037	0.1003	0.1269	0.1037
QQQ	15m	0.0870	0.0825	0.0879	0.0857	0.0825
QQQ	30m	0.0687	0.0685	0.0688	0.0686	0.0685
IWM	15s	0.2007	0.0516	0.0474	0.2033	0.1599
IWM	30s	0.1858	0.0672	0.0623	0.1851	0.1553
IWM	1m	0.1699	0.0915	0.0848	0.1715	0.1565
IWM	5m	0.1258	0.1206	0.1219	0.1271	0.1271
IWM	15m	0.0825	0.0794	0.0830	0.0820	0.0822
IWM	30m	0.0697	0.0686	0.0689	0.0708	0.0703
EFA	15s	0.1063	0.0713	0.0792	0.1093	0.0911
EFA	30s	0.1182	0.1063	0.1022	0.1168	0.1195
EFA	1m	0.1068	0.1035	0.0964	0.1052	0.1105
EFA	5m	0.0935	0.0928	0.0919	0.0929	0.0934
EFA	15m	0.0732	0.0718	0.0720	0.0716	0.0726
EFA	30m	0.0779	0.0779	0.0763	0.0790	0.0799
TLT	15s	0.0605	0.0669	0.0700	0.0602	0.0604
TLT	30s	0.0736	0.0688	0.0734	0.0704	0.0630
TLT	1m	0.0548	0.0621	0.0649	0.0531	0.0557
TLT	5m	0.0453	0.0484	0.0474	0.0468	0.0486
TLT	15m	0.0443	0.0468	0.0451	0.0466	0.0476
TLT	30m	0.0266	0.0289	0.0267	0.0294	0.0296

The complete supplementary tables are `results/etf_breadth_incidence.csv`, `results/etf_breadth_H.csv`, `results/etf_breadth_risk.csv` and `results/etf_breadth_intervals.csv`. They include bands 1–5 and 1–22, the original fixed risk rule for all constructions/grids, and all registered mean-block 22/11/44 intervals. These tables are aggregate research summaries; licensed daily observations are excluded. A13 follows the original result and is not untouched validation. The original A11 policy grid remains explicitly post-result sensitivity; no new preferred policy is selected from this extension.